

ITA0608 - Machine Learning

List of Lab Exercises

20. To predict future sales using the Linear Regression algorithm.

Aim:

To predict future sales using the Linear Regression algorithm.

Algorithm:

- 1.Import the required Python libraries and Linear Regression model.
- 2.Create the sales dataset with input features and sales values.
- 3.Separate the input features (X) and target variable (y).
- 4.Train the Linear Regression model using the dataset.
- 5.Provide the future time period or input value as a new sample.
- 6.Predict the future sales using the trained model.
- 7.Display the predicted sales result.

Dataset:

Program Code:

```
import pandas as pd

from sklearn.linear_model import LinearRegression

data = {

    'AdvertisingBudget': [20,25,30,35,40,45,50,55,60,65],

    'Salespersons': [2,2,3,3,4,4,5,5,6,6],

    'MonthlySales': [250,290,340,390,450,510,580,640,710,780]

}
```

```
df = pd.DataFrame(data)

X = df[['AdvertisingBudget', 'Salespersons']]

y = df['MonthlySales']

model = LinearRegression()

model.fit(X, y)

new_sample = [[48, 5]]

prediction = model.predict(new_sample)

print("Predicted Monthly Sales:", round(prediction[0], 2), "Units")
```

Output:

```
model.fit(X, y)

# New sample
new_sample = [[48, 5]]

# Prediction
prediction = model.predict(new_sample)

print("Predicted Monthly Sales:", round(prediction[0], 2), "Units")

... Predicted Monthly Sales: 563.1 Units
```

Dataset 2:

Program Code:

```
import pandas as pd

from sklearn.linear_model import LinearRegression

data = {

    'StoreArea': [1000,1200,1400,1600,1800,2000,2200,2400,2600,2800],

    'DailyCustomers': [120,150,180,220,260,300,340,380,420,460],
```

```
'MonthlySales': [8,10,12,15,18,22,26,31,36,42]

}
```

```
df = pd.DataFrame(data)
```

```
X = df[['StoreArea', 'DailyCustomers']]
```

```
y = df['MonthlySales']
```

```
model = LinearRegression()
```

```
model.fit(X, y)
```

```
new_sample = [[2100, 320]]
```

```
prediction = model.predict(new_sample)
```

```
print("Predicted Monthly Sales:", round(prediction[0], 2), "₹ Lakhs")
```

Output:

```
# Input features and target
X = df[['StoreArea', 'DailyCustomers']]
y = df['MonthlySales']

# Train model
model = LinearRegression()
model.fit(X, y)

# New sample
new_sample = [[2100, 320]]

# Prediction
prediction = model.predict(new_sample)

print("Predicted Monthly Sales:", round(prediction[0], 2), "₹ Lakhs")
```

```
... Predicted Monthly Sales: 25.29 ₹ Lakhs
```

Dataset 3:

Program Code:

```
import pandas as pd

from sklearn.linear_model import LinearRegression

data = {

    'WebsiteVisitors': [1000,1500,2000,2500,3000,3500,4000,4500,5000,5500],

    'AdvertisementCost': [5000,7000,9000,11000,13000,15000,17000,19000,21000,23000],

    'SalesOrders': [120,180,250,320,400,490,590,700,820,950]

}


df = pd.DataFrame(data)

X = df[['WebsiteVisitors', 'AdvertisementCost']]

y = df['SalesOrders']

model = LinearRegression()

model.fit(X, y)

new_sample = [[4200, 18000]]

prediction = model.predict(new_sample)

print("Predicted Sales:", round(prediction[0], 2), "Orders")
```

Output:

```
model.fit(X, y)

# New sample
new_sample = [[4200, 18000]]

# Prediction
prediction = model.predict(new_sample)

print("Predicted Sales:", round(prediction[0], 2), "Orders")

*** Predicted Sales: 664.49 Orders
```

Dataset 4:

Program Code:

```
import pandas as pd

from sklearn.linear_model import LinearRegression

data = {

    'Discount': [5,8,10,12,15,18,20,22,25,28],

    'CustomersPerDay': [150,180,220,260,300,340,380,420,460,500],

    'WeeklySales': [220,260,310,360,420,490,570,650,740,840]

}

df = pd.DataFrame(data)

X = df[['Discount', 'CustomersPerDay']]

y = df['WeeklySales']

model = LinearRegression()

model.fit(X, y)
```

```
new_sample = [[18, 350]]
```

```
prediction = model.predict(new_sample)
```

```
print("Predicted Weekly Sales:", round(prediction[0], 2), "₹ Thousands")
```

Output:

```
# Train model
model = LinearRegression()
model.fit(X, y)

# New sample
new_sample = [[18, 350]]

# Prediction
prediction = model.predict(new_sample)

print("Predicted Weekly Sales:", round(prediction[0], 2), "₹ Thousands")

*** Predicted Weekly Sales: 535.66 ₹ Thousands
```

Dataset 5:

Program Code:

```
import pandas as pd
```

```
from sklearn.linear_model import LinearRegression
```

```
data = {
```

```
    'MarketingBudget': [30,35,40,45,50,55,60,65,70,75],
```

```
    'TestDrives': [20,25,30,35,40,45,50,55,60,65],
```

```
    'CarsSold': [8,10,13,17,22,28,35,43,52,62]
```

```
}
```

```
df = pd.DataFrame(data)
```

```
X = df[['MarketingBudget', 'TestDrives']]

y = df['CarsSold']

model = LinearRegression()

model.fit(X, y)

new_sample = [[58, 48]]

prediction = model.predict(new_sample)

print("Predicted Cars Sold:", round(prediction[0], 2))
```

Output:

```
y = df['CarsSold']

# Train model
model = LinearRegression()
model.fit(X, y)

# New sample
new_sample = [[58, 48]]

# Prediction
prediction = model.predict(new_sample)

print("Predicted Cars Sold:", round(prediction[0], 2))
```

```
... Predicted Cars Sold: 35.6
```

Result:

The Linear Regression model successfully predicted future sales based on the given input features.

The predicted sales value was displayed for the new sample, demonstrating the model's ability to estimate future sales.

